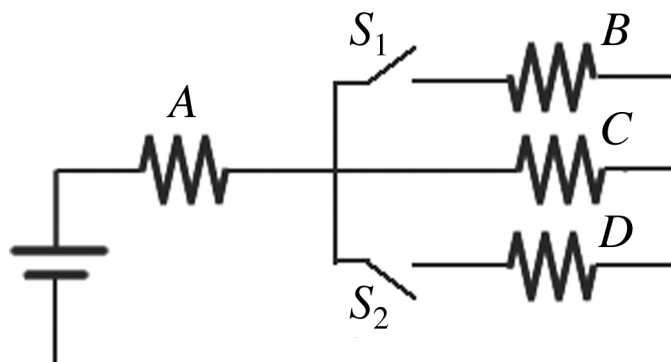
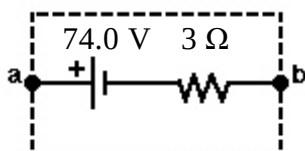


MULTIPLE CHOICE. Choose the one alternative that best completes the statement or answers the question.

- 1) In the circuit shown in the figure, four identical resistors labeled *A* to *D* are connected to a battery as shown. S_1 and S_2 are switches. Which of the following actions would result in the GREATEST amount of current through resistor *A*? 1) _____



- A) leaving both switches open as shown. B) closing S_1 only
C) closing both switches D) closing S_2 only
- 2) If a light bulb uses 0.5A of current, how much charge flows in it in 10 minutes ? 2) _____
A) 30C B) 300 C C) 6C D) 5C
- 3) The rms voltage from the grid (wall outlets) is: 3) _____
A) 120V B) 1.5V C) 12V D) 9V
- 4) If your electric iron uses 1000 W on a 115 V line, how much current does it draw? 4) _____
(about)
A) 115 mA B) 8.5 A C) 10A D) 13A
- 5) The voltage of a car battery is 5) _____
A) 1.5V B) 12V C) 120V D) 9V
- 6) Figure 19.9a 6) _____



The emf and the internal resistance of a battery are as shown. In Figure 19.9a, a current of 6.4 A is drawn from the battery when a resistor is connected across the terminals.

The power dissipated **by the resistor** is closest to:

- A) 470 W B) 410 W C) 350 W D) 380 W E) 440 W

- 7) resistors are placed in parallel. The effective resistance is: 7) _____
- A) equal to the smallest resistance
 - B) equal to the largest resistance
 - C) the sum of the individual resistance
 - D) smaller than the smallest of the resistances

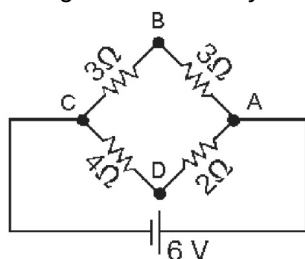
- 8) A certain microwave oven draws 10.0 A of current when it is operated at 120 V household lines. If electrical energy costs 10 cents per kilowatt-hour, how much does it cost to operate the microwave oven for **30 minutes** every day for 1 month (30 days) ? (in cents) hint: 30min is 0.5 hours 8) _____

MULTIPLE CHOICE. Choose the one alternative that best completes the statement or answers the question.

- 9) 3 resistors are in series with a power supply. 5 ohms. 20 ohms. 25 ohms . The power supply delivers 50V. What is the voltage across the 20 ohms resistance? 9) _____
- A) 10V
 - B) 50V
 - C) 20V
 - D) 25V

- 10) A coulomb of charge flowing in a bulb filament powered by a 6-volt battery is provided with 10) _____
- A) 6 newtons.
 - B) 6 watts.
 - C) 6 ohms.
 - D) 6 amperes.
 - E) 6 joules.

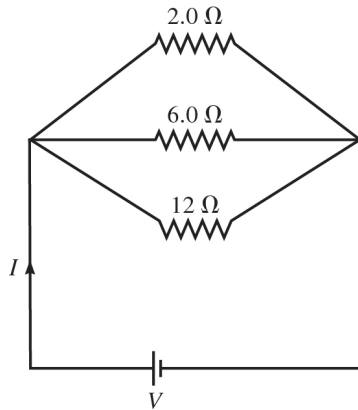
- 11) What is the magnitude of the potential difference between points C and D for the circuit shown in the figure? The battery is ideal, and all the numbers are accurate to two significant figures. 11) _____



- A) 6.0 V B) 4.0 V C) 3.0 V D) 2.0 V
- 12) To produce 400 joules of heat every second (J/s) a 120-V line, a heating element should have a resistance of ? 12) _____
- A) 36 ohms
 - B) 120 ohms
 - C) 0.3 ohms
 - D) 3.33 ohms
- 13) A charge of 12 C passes through an electroplating apparatus in 2.0 min. What is the average current in the apparatus? 13) _____
- A) 0.10 A
 - B) 0.60 A
 - C) 6.0 A
 - D) 1.0 A

14) An ampere can be defined as: 14) _____
 A) Ω/V B) N/C C) C/s D) J/C

15) Three resistors with resistances of $2.0\ \Omega$, $6.0\ \Omega$, and $12\ \Omega$ are connected across an ideal dc voltage source V as shown in the figure. If the total current through the circuit is $I = 2.0\ A$, what is the applied voltage V ? 15) _____



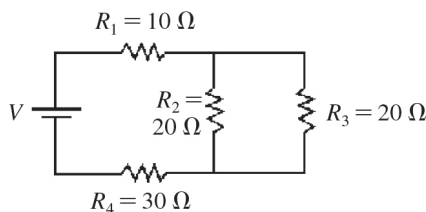
A) $6.0\ V$ B) $2.0\ V$ C) $2.7\ V$ D) $1.5\ V$ E) $3.0\ V$

16) The rate at which energy is converted from one form to another is measured in 16) _____
 A) amperes B) volts C) watts D) joules

17) The voltage from the grid in Europe has an amplitude of $300\ V$ and a frequency of $50\ Hz$. Compute the period T of the signal. 17) _____
 A) $0.02s$ B) $6s$ C) $50s$ D) $0.17s$

18) A $100\text{-}W$ lamp glows brighter than a $25\text{-}W$ lamp. The electrical resistance of the $100\text{-}W$ lamp is 18) _____
 A) greater. B) less. C) the same.

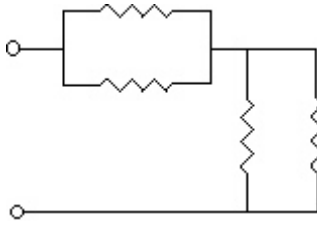
19) What is the equivalent resistance in the circuit shown in the figure? 19) _____



A) $35\ \Omega$ B) $50\ \Omega$ C) $55\ \Omega$ D) $80\ \Omega$

- 20) The resistors in the circuit shown in Figure 19.2 each have a resistance of $700\ \Omega$. What is the equivalent resistance of the circuit? 20) _____

Figure 19.2



- A) $1,400\ \Omega$ B) $700\ \Omega$ C) $175\ \Omega$ D) $2,800\ \Omega$
- 21) This is not a questions. These are equations. 21) _____

$$\Delta V = \frac{Q}{C}$$

$$I = \frac{\Delta Q}{\Delta t}$$

$$R = \frac{\rho \ell}{A}$$

$$P = I \Delta V$$

$$I = \frac{\Delta V}{R}$$

$$R_s = \sum_i R_i$$

$$\frac{1}{R_p} = \sum_i \frac{1}{R_i}$$

Answer Key

Testname: TEST1

- 1) C
- 2) B
- 3) A
- 4) B
- 5) B
- 6) C
- 7) D
- 8)
- 9) C
- 10) E
- 11) B
- 12) A
- 13) A
- 14) C
- 15) C
- 16) C
- 17) A
- 18) B
- 19) B
- 20) B
- 21)